

Learning and Remembering Command Names

John B. Black
Yale University
Department of Psychology
New Haven, CT 06520

Thomas P. Moran
Xerox Palo Alto Research Center
3333 Coyote Hill Road
Palo Alto, CA 94304

Natural language would seem to have a strong effect on users' behavior with artificial command languages for interacting with computer systems. For example, Ledgard, Whiteside, Singer, and Seymour [7] claim to have shown that user performance on a text-editing system is better when the command language for the system uses natural language terminology than when it does not. Although the Ledgard et al. data show an overall superiority for the "natural language" version of their editor, there were so many differences between the two system versions in their study that we cannot possibly know which of the differences were critical. Natural language is not a monolithic entity. Thus, there needs to be more specific, controlled experiments to determine which aspects of natural language facilitate human-computer interaction.

We can divide the potential effects of natural language on command languages into: (a) effects on the names of commands, (b) effects on command arguments, and (c) effects on how command-argument units are interrelated (see Black and Sebrechts [2]). Others have investigated arguments (Barnard et al. [1]) and command-argument interrelations (Carroll [4]). In this paper, we describe our research concerning the first of these—namely, how the names of commands effect the learnability and memorability of the commands. Our investigation uses text-editing as the specific domain.

Applied research in human-computer interaction is a subtle affair, with many pitfalls awaiting the unwary researcher. Thus, in addition to presenting research results, we will conclude this paper with some methodological lessons.

DESCRIPTIVE TERMS GENERATED BY NON-COMPUTER-USERS

Initially, we tried a straightforward method to determine good names for editing commands—simply observe how non-computer-users naturally describe editing operations (changes). We gave a group of subjects a marked-up manuscript indicating text-editing operations and asked the subjects to write one sentence descriptions of the operations. The results of this initial study proved unsatisfactory in two ways. First, many of the descriptions were at a higher level of generality than we wanted. For example, when presented with the text *the reded car* with an X through the second *ed*, the subjects would describe it as "correct the spelling of the second word." This description gives the goal of the operation rather than the actual operation on the text. We instructed later subjects not to give such responses, but they then tended to give descriptions of the edit marks on the manuscript rather than the operations represented by the marks. For instance, a typical description of the example above was "crossout the last two letters in the second word." Here, the subject's use of the verb *crossout* is clearly influenced by the particular mark we used to designate the operation.

We solved the problem of notational-influence by presenting editing tasks to subjects as *before-after pairs*, i.e., showing the state of the text before and after an editing operation. For example, a typical before-after pair was:

Before: witout doing
After: without doing

The subjects described INSERT operations like this one using the verb *add*, DELETE operations using the verb *omit*, REPLACE operations using *change*, and MOVE operations using *put*. Typically, computer text editors use the names *insert*, *delete*, *replace*, and *move* for these operations; so the subjects' descriptive verbs were quite different from those generated by system designers.

We conducted a preliminary experiment to see if the

subject-generated verbs were more memorable as command names than the designer-generated names. Groups of subjects learned to apply the command names to a set of before-after pairs. A day later, we tested their memory for these names and how to use them. One group of subjects learned the designer-generated names (*insert*, *delete*, *replace*, and *move*) and the other group learned the subject-generated names (*add*, *omit*, *change*, and *put*). The memory results showed no difference between subject-generated and designer-generated command names.

Interesting. But even if we had found a difference, "subject-generated" vs. "designer-generated" names would not have been a very helpful distinction. This experiment would not have revealed the underlying psychological factors (Moran [8]) that make names learnable and memorable.

FACTORS AFFECTING LEARNING AND MEMORY

What might the underlying factors be? In the preliminary study, we noticed two differences between the subject-generated names and the typical command names—frequency and generality. The subject-generated names tended to be words that are more frequent in written English (as measured by the Kucera and Francis [7] word frequency counts). For example, the subject-generated names *add* and *omit* occur much more frequently in English than do the typical names *insert* and *delete*; so the average person would have had more experience with the subject-generated names. (Of course, *insert* and *delete* are very frequent and familiar terms to people associated with computers.) Thus, frequency could be an important factor for command names.

Another feature of the subject-generated names was that they were general terms. For example, the subject-generated term *put* was intended to refer to the MOVE operation, but it could just as well refer to the INSERT or REPLACE operations. The typical term *move*, however, could only refer to the movement operation. We postulate that the important underlying factor here is the *discriminability* of the name. *Move* is more discriminating than *put*, since it discriminates more sharply which operation it refers to within the set of operations. In addition to being too general, there is another way for a name to be nondiscriminating—it can refer to no operations at all (i.e., the association between the name and the operation is totally arbitrary and non-mnemonic). For example, the term *walk* is nondiscriminating among text-editing operations. We would expect that the less discriminating a name is, the harder it should be to learn and remember.

Finally, we have been presupposing that command names are English words. Clearly, they do not have to be. Nonsense syllables (a venerable psychological tradition) and icons (see Hemenway [5]), for example, could be command names. We might expect words to be better because of the rich semantics associated with them, but this is another factor deserving investigation.

A COMMAND NAMING EXPERIMENT

We conducted an experiment that systematically varied these factors (frequency, discriminability, and word/nonword). The experiment had seven naming conditions, with different groups of 12 subjects tested in each condition (i.e., 84 subjects in all). Each condition consisted of a set of eight command names for an eight-operation text-editing system. The eight operations were INSERT, DELETE, REPLACE, MOVE, TRANSPOSE, CAPITALIZE, ERASE (delete but don't close up the space), and SPACE (insert space). The seven conditions, along with examples of the names used for the INSERT and DELETE operations, were as follows:

Infrequent, discriminating words	<i>insert</i>	<i>delete</i>
Frequent, discriminating words	<i>add</i>	<i>remove</i>
Infrequent, nondiscriminating words	<i>amble</i>	<i>perceive</i>
Frequent, nondiscriminating words	<i>walk</i>	<i>view</i>
General (frequent, nondiscriminating) words	<i>alter</i>	<i>correct</i>
Nondiscriminating nonwords (nonsense)	<i>GAC</i>	<i>MIK</i>
Discriminating nonwords (icons)	<i>abc-adbc</i>	<i>abc-ac</i>

Thus, there were four discriminating name conditions and three nondiscriminating name conditions, five word conditions and two nonword conditions. Perhaps the most unusual condition was the discriminating nonwords, which symbolized the operations with examples of before-after pairs; hence, we called these *icons*.

The procedure for the experiment had two phases—learning and memory. The learning phase had two parts—an introduction and a sequence of training trials. In the introduction, the subjects were given an explanation of the task that included examples of the materials to be used. The subjects were then given decks of cards containing the training materials. Each card contained a before-after text pair on the front and the name of the appropriate command on the back. The subjects wrote down the name they thought applied to the pair, then turned over the card to discover the correct answer. The cards were presented in blocks of 16 before-after pairs, each block constituting one learning trial. There were 11 trials in all. The subjects returned a week later for the memory phase of the experiment, which had two parts. First, the subjects were told to recall, in any

order, the eight names that they had learned. The subjects were told the eight names after finishing the free recall. Then they were given a list of 32 before-after pairs and told to write down the applicable name for each pair (without feedback).

Thus, there were three kinds of tasks on which the subjects' performance was measured: (a) the number of trials to reach a criterion level of learning (arbitrarily set at 80% correct), (b) the percentage of errors in freely recalling the command names, and (c) the percentage of errors in choosing the command names when cued with before-after pairs. The performance data on the three tasks are plotted in the three parts of Figure 1. The following are the salient results:

(1) On all three tasks, the performance was best with the infrequent, discriminating words: these command names were learned the fastest (Figure 1a), recalled the best (Figure 1b), and most accurately chosen to match before-after pairs (Figure 1c). The general words performed worst on learning and cued choice.

(2) Overall, discriminating words performed better than nondiscriminating words on initial learning and on later choice. (2a) However, discriminability had no effect in free recall. Since the referents of the names (i.e., the editing operations in the form of before-after pairs) are not present during the free recall task, the discriminability of names has no effect. (2b) Also, discriminability has no effect with nonwords. In fact, the icons (discriminable nonwords) were harder to learn than the nonsense syllables (nondiscriminable nonwords). This learning difficulty with the icons is at present a mystery to us.

(3) Words were always better than nonwords in free recall, but they were better in learning and cued choice only when they were discriminating. In fact, learning and cued choice performance on nondiscriminating words was worse than nonwords.

(4) Word frequency also had an effect, with the infrequent words performing better in all tasks. However, the effects of word frequency interacted with discriminability in learning and cued choice (the latter perhaps due to a ceiling effect).

We do not report a full analysis of this data in this short paper. We also have detailed data on the specific name confusions made by the subjects during learning and cued choice. Further, we are attempting to explain this data with a model based on semantic features. In this kind of model, the semantics of each name and operation is represented as a set of features, and name choices and confusions are predicted by simple matching functions on feature sets.

METHODOLOGICAL LESSONS

These studies illustrate a number of general lessons about the conduct of research on human-computer interaction:

(1) A computer system cannot be designed by simply asking computer-naïve people what they think it should be like; they are not good designers. As we saw above, the names people spontaneously use to describe editing operations cannot be assumed to be good command names. This conclusion is reinforced by the similar findings of Barnard et al. [1] and Carroll [4].

(2) Researchers must be careful to use representative tasks on which to base their conclusions about systems. For example, the pattern of results in the free recall task (Figure 1b) was quite different than the pattern of results in the learning and cued choice tasks (Figure 1a and 1c). If we had used only the simple free recall task, then our conclusions would have been distorted (e.g., we would have concluded that the discriminability of names had no effect). This distortion would have been especially serious, because this task is the least representative of what people do when they use systems. The experimental procedure we have chosen above was designed to be a reasonable simplification of real system learning and usage situations. The design was a compromise between being a good simplification and experimental control and convenience (e.g., the experiments were run with pencil-and-paper in a classroom setting). Eventually, the results from such experiments must be checked in more realistic situations.

(3) While exploratory studies of global effects are worthwhile, we must not stop there. We need to probe the underlying psychological factors that produce the effects. For example, from a general assertion that use of natural language terminology facilitates user performance, we might conclude that it is always better to use words than nonwords. However, the results of our experiment have shown that this is not the case. In particular, discriminating words facilitate user performance, but nondiscriminating ones actually yield worse performance than nonwords.

(4) We would push this argument even further. Beyond factors, what we would really like to have are specific models for predicting user performance, for such models are the most effective way to transmit research results to system designers (Moran [8]; Card, Moran, and Newell [3]). In the case of command naming, we propose that a semantic feature model is a promising candidate for evaluating how well specific command names work. See the accompanying paper by Rosenberg [9] for an example

of the kind of model we have in mind. Of course, such models are difficult to formulate. Our research on naming has not yet reached this stage of refinement—it is still at the stage of initial factor exploration—but this is the direction that we would have it go.

REFERENCES

- [1] Barnard, P. J., Hammond, N. V., Morton, J., Long, J. B., and Clark, I. A. Consistency and compatibility in human-computer dialogue. *International Journal of Man-Machine Studies*, 1981, 15, 87-134.
- [2] Black, J. B., and Sebrechts, M. M. Facilitating human-computer communication. *Applied Psycholinguistics*, 1981, 2, 146-177.
- [3] Card, S. K., Moran, T. P., and Newell A. *The Psychology of Human-Computer Interaction*. Hillsdale, N.J.: Erlbaum, 1982.
- [4] Carroll, J. M. Learning, using, and designing command paradigms. *IBM Research Report RC 8141*, 1980.
- [5] Hemenway, K. Psychological issues in the use of icons in command names. *Conference on Human Factors in Computer Systems*, Gaithersburg, Md., March 1982.
- [6] Kucera, H., and Francis, W. N. *Computational Analysis of Present Day American English*. Providence, R.I.: Brown University Press, 1967.
- [7] Ledgard, H., Whiteside, J. A., Singer, A., and Seymour, W. The natural language of interactive systems. *Communications of the ACM*, 1980, 10, 556-563.
- [8] Moran, T. P. An applied psychology of the user. *Computing Surveys*, 1981, 13, 1-11.
- [9] Rosenberg, J. K. Evaluation of the suggestiveness of command names. *Conference on Human Factors in Computer Systems*, Gaithersburg, Md., March 1982.

FIGURE 1. RESULTS OF COMMAND NAMING EXPERIMENT

